

The University of Melbourne
School of Computing and Information Systems
COMP90086 Computer Vision, 2025 Semester 2

Assignment 3: Autoencoders and Semi-supervised learning

Due: 23:59, 12 Oct 2025
Submission: Source code (in Jupyter Notebook) and written responses (as .pdf)
Marks: The assignment will be marked out of 7 points, and will contribute 7% of your total mark.

A common problem in machine learning for computer vision is the cost of obtaining a large body of labelled data for training an image classifier. In this assignment, you will **explore the use of autoencoders to build representations from a large set of *unlabelled* data**, and then use those representations to train a classifier from a small set of *labelled* data. This process is often called semi-supervised learning. You will design and improve algorithms for the autoencoder and classifier so that you maximise the performance of the classifier when trained on a small set of labelled data.

The assignment uses the MNIST data set of hand written digits. In this assignment, you will use 80% of the MNIST training set (with 20% reserved for validation) to train your autoencoder. You will then use just 0.1% (just 60 examples) to train a classifier using the representations created by the autoencoder. You will also train a conventional classifier using the same tiny training set and observe how the autoencoder-based classifier compares to the conventional classifier. These settings (80% and 0.1%) are written in the code in the jupyter notebook you will download as the basis for your assignment.

1. Existing Code: reconstruction and classification error analysis [2 marks]

Run the existing Jupyter Notebook and **observe how well the autoencoder works (how good are the reconstructions and the accuracy of the classifier based on the autoencoder representations and compare this to the baseline conventional classifier provided)**. Your report should **describe the encoder, decoder and classifier and explain how they interact and how this design uses the unlabelled data to help a classifier that is trained on the small labelled data set**.

You should give **quantitative results for the networks** and describe any observations and conclusions you can draw from their accuracy and loss training graphs.

2. Varying the size of the latent representation [2 marks]

The sample code uses a latent representation size of 16 dimensions. You should explore the consequences of varying this in terms of both the autoencoder reconstruction accuracy and the classifier accuracy. Your search should be guided by finding the latent representation size that is best for this task.

In your written report, you should report the results for at least 5 latent sizes, give the best size and explain your search strategy.

3. Autoencoder and Classifier design [3 marks]

Modify the design of the autoencoder and classifier designs and try to improve the performance of your system. Some things to consider include:

- Try different autoencoder architectures
- Try different classifier architectures
- Try finding the best latent size for your design (this might not be the same as the best size for the design provided in the jupyter notebook)
- You may like to try increasing the learning rate to speed up your design cycle if training is taking a long time.

In your written report, describe your design choices and explain why you chose them and how you tested their effectiveness. You should also give the performance of each design.

Submission

You should make two submissions on the LMS: your code and a short written report explaining your method and results. **The response to each question should be no more than 400 words.** Submission will be made via the Canvas LMS. Please submit your code and written report separately under the Assignment 3: Code and the Assignment 3: Report links on Canvas.

- Your **code** submission should include the Jupyter Notebook (please use the provided template) with your code and any additional files we will need to run your code, if any (do not include any image files).
- Your **written report** should be a .pdf with your answers to each of the questions. The report should address the questions posed in this assignment and include any images, diagrams, or tables required by the question.

Evaluation

Your submission will be marked on the correctness of your code/method, including the quality and efficiency of your code and the appropriateness of design decisions. You should

use built-in Python functions where appropriate and use descriptive variable names. Your written report should clearly explain your approach and any experimentation used to produce your results, and include all of the specific outputs required by the question (e.g., images, diagrams, tables, or responses to sub-questions).

Late submission

The submission mechanism will stay open for one week after the submission deadline. Late submissions will be penalised at 10% of the total possible mark per 24-hour period after the original deadline. Submissions will be closed 7 days (168 hours) after the published assignment deadline, and no further submissions will be accepted after this point.

To request an extension on this assignment, please see the [FEIT extension policy](#) and follow the steps below:

- **To request an extension of 1-3 days (24-72 hours)**, complete the online declaration form available at the link above.
- **To request an extension of 4-7 days**, please apply for Special Consideration.

Please note that extension requests should be made before the assignment is due. Late extension requests can only be granted through Special Consideration. The longest possible extension for this assignment is 7 days (168 hours).

Updates to the assignment specifications

If any changes or clarifications are made to the project specification, these will be posted on the LMS.

Academic misconduct

You are welcome — indeed encouraged — to collaborate with your peers in terms of the conceptualisation and framing of the problem. For example, we encourage you to discuss what the assignment specification is asking you to do, or what you would need to implement to be able to respond to a question.

However, sharing materials — for example, showing other students your code or colluding in writing responses to questions — or plagiarising existing code or material (including materials generated by chatGPT or other AI) will be considered cheating. Your submission must be your own original, individual work. We will invoke University's [Academic Misconduct policy](#) where inappropriate levels of plagiarism or collusion are deemed to have taken place.